SCS CVX Julia DCP - disciplined convex programming SOCP
Norm approximation

1 Norm Approximation

Let $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{b} \in \mathbb{R}^m$. The objective is to minimize the residual vector $\mathbf{r} = \mathbf{A}\mathbf{x} - \mathbf{b}$.

The general problem is formulated as,

$$\min_{\mathbf{x}\in\mathbb{R}^n} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|. \tag{1}$$

Depending on the norm chosen, this results in different convex optimization problems with unique statistical interpretations and solution characteristics.

1.0.1 ℓ_1 Norm Approximation (Sum of Absolute Residuals)

$$\|Ax - b\|_1 = \sum_{i=1}^{m} |(Ax - b)_i| \tag{2}$$

Also known as **Least Absolute Deviations (LAD)**.

- **Robustness:** Unlike the ℓ_2 norm, the ℓ_1 norm places less weight on large residuals (outliers). It yields a robust estimator where the solution tends to minimize the median of the residuals rather than the mean.
- **Sparsity:** In problems like Basis Pursuit (where we minimize $\|x\|_1$ subject to constraints), ℓ_1 induces sparsity. Here, in residual minimization, it allows many residual entries to be exactly zero.

Convex Formulation (Linear Program): The problem is non-differentiable at zero. We reformulate it as a Linear Program (LP) by introducing slack variables $t \in \mathbb{R}^m$:

$$\begin{array}{ll} \text{minimize} & \mathbf{1}^T t \\ \text{subject to} & -t \preceq Ax - b \preceq t \end{array} \tag{3}$$

where $\preceq$ denotes component-wise inequality.

1.0.2 ℓ_2 Norm Approximation (Least Squares)

$$\|Ax - b\|_2 = \left(\sum_{i=1}^{m} (Ax - b)_i^2\right)^{1/2} \tag{4}$$

Minimizing the ℓ_2 norm is equivalent to minimizing the squared Euclidean norm $\|Ax - b\|_2^2$. This is the standard **Least Squares** problem.

- **Statistical Interpretation:** Corresponds to the Maximum Likelihood Estimator (MLE) under the assumption of Gaussian noise.
- **Sensitivity:** Highly sensitive to outliers because errors are squared.

Convex Formulation (Unconstrained Quadratic): This is a differentiable, unconstrained convex problem with a closed-form analytical solution (assuming A is full rank):

$$x^\star = (A^T A)^{-1} A^T b \tag{5}$$

1.0.3 ℓ_∞ Norm Approximation (Chebyshev)

$$\|Ax - b\|_\infty = \max_{1 \le i \le m} |(Ax - b)_i| \tag{6}$$

Also known as **Minimax** or **Chebyshev approximation**.

- **Goal:** Minimizes the worst-case residual.
- **Applications:** Critical in control design and engineering where ensuring that error never exceeds a safety threshold is more important than the average error.

Convex Formulation (Linear Program): We introduce a scalar variable $t \in \mathbb{R}$ representing the maximum bound:

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & -t\mathbf{1} \preceq Ax - b \preceq t\mathbf{1} \end{array} \tag{7}$$

1.1 Penalty Functions

In many approximation problems, we minimize a sum of penalty functions applied to the residuals $r_i = (Ax - b)_i$. The problem takes the form:

$$\min_x \sum_{i=1}^m \phi(r_i) \tag{8}$$

where $\phi : \mathbb{R} \to \mathbb{R}$ is a convex penalty function.

1.1.1 Norm Based ($p = 1, 2, \infty$)

$\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_\infty$.

1.1.2 Dead Zone Linear

Functions of the form, for $r \in \mathbb{R}^n$,

$$\sum_i \phi(r_i), \tag{9}$$

where

$$\phi(r) = \max\{|r| - a, 0\}. \tag{10}$$

1.1.3 Log Barrier

$$\phi(u) = \begin{cases} -\log(1 - u^2), & |u| < 1 \\ \infty, & |u| \ge 1 \end{cases} \tag{11}$$

1.1.4 Huber Penalty

2 Moreau Envelope

Definition 2.1 (Moreau Envelope): *Given $f : \mathbb{R}^n \to \mathbb{R}$, we have the Moreau Envelope as*

$$f_u(\mathbf{x}) = \inf_{\mathbf{y}\in\mathbb{R}^n} \left\{ f(\mathbf{y}) + \frac{1}{2u}\|\mathbf{x}-\mathbf{y}\|^2 \right\}.$$

Remark 2.2: *The above expression is essentially an infimal convolution of $uf : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$ where $g = (1/2)\|\cdot\|_2^2$.*

Definition 2.3 (Proximal Operator): *The proximal operator $prox_u(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}^n$ of f with parameter u is defined as,*

$$prox_u(\mathbf{x}) = \arg\min_{\mathbf{y}\in\mathbb{R}^n} \left\{ f(\mathbf{y}) + \frac{1}{2u}\|\mathbf{x}-\mathbf{y}\|^2 \right\}.$$

2.0.1 Properties of Moreau Envelope

1. $f_u(\mathbf{x})$ is differentiable if f is continuous. Further, the derivative is continuous.
2. $f_u(\mathbf{x})$ is defined for all $\mathbf{x} \in \mathbb{R}^n$, even when f is not.
3. $f_u(\mathbf{x})$ is convex whenever f is convex.
4. If x^* is the unique minimizer of f, then it is the unique minimizer of $f_u(\mathbf{x})$

Remark 2.4: *$f_u(\mathbf{x})$ is convex, since it is the min over one variable of a jointly convex function.*

Remark 2.5: *$f_u(\mathbf{x})$ is continuously smooth.*

Remark 2.6:

$$epi f_u(x) = epi f(x) \oplus epi \left(\frac{1}{2u}\|x\|^2\right)$$

Remark 2.7: *x^* is a minima of f if and only if x^* is a minima of f_u.*

Remark 2.8:

$$\nabla f_u(\mathbf{x}) = \frac{1}{u}(x - prox_u(x)).$$

3 Epigraphs

TODO: complete this
Duality
Supporting Hyperplane Theorem

Definition 3.1 (Conjugate of a function): *Let $f : \mathbb{R}^n \to \mathbb{R}$. The conjugate $f^* : \mathbb{R}^n \to \mathbb{R}$ of f is defined as,*

$$f^*(\mathbf{y}) = \sup_{\mathbf{x}\in dom f} (\mathbf{y}^T\mathbf{x} - f(\mathbf{x})). \tag{12}$$

Remark 3.2: *The conjugate of f is also called Legendre transform, polar of f, Fenchel dual.*

Remark 3.3: *f^* is convex, irrespective of f.*

Remark 3.4:

$$-f^*(0) = \min_{\mathbf{x}\in\mathbb{R}^n} f(\mathbf{x}) \tag{13}$$

Theorem 3.5 (Fenchel Young inequality):

$$f(\mathbf{x}) + f^*(\mathbf{m}) \geq \mathbf{m}^T\mathbf{x} \tag{14}$$

3.1 Recovering f from f^*

Definition 3.6 (Biconjugate): *Let $f^* : \mathbb{R}^n \to \mathbb{R}\cup\{+\infty\}$ be the conjugate of a function $f : \mathbb{R}^n \to \mathbb{R}\cup\{+\infty\}$. The **biconjugate** $f^{**} : \mathbb{R}^n \to \mathbb{R}\cup\{+\infty\}$ is defined as the conjugate of f^*,*

$$f^{**}(\mathbf{x}) = (f^*)^*(\mathbf{x}) = \sup_{\mathbf{y}\in\mathbb{R}^n} \left(\mathbf{y}^T\mathbf{x} - f^*(\mathbf{y})\right). \tag{15}$$

Remark 3.7 (Geometric Interpretation): *The biconjugate f^{**} constructs the pointwise supremum (upper envelope) of all affine minorants of f. Specifically, the epigraph of f^{**} is the **closed convex hull** of the epigraph of f,*

$$epi\, f^{**} = cl(conv(epi\, f)). \tag{16}$$

Thus, f^{**} represents the "tightest" closed convex function that minorizes f.

3.1.1 Properties of the Biconjugate

Corollary 3.8 (Weak Duality): *For any proper function f, the inequality $f^{**}(\mathbf{x}) \leq f(\mathbf{x})$ holds for all $\mathbf{x}\in\mathbb{R}^n$.*

Proof. Recall the **Fenchel-Young inequality**, which states that for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$f^*(\mathbf{y}) + f(\mathbf{x}) \geq \mathbf{x}^T\mathbf{y}. \tag{17}$$

Rearranging terms yields:

$$\mathbf{x}^T\mathbf{y} - f^*(\mathbf{y}) \leq f(\mathbf{x}). \tag{18}$$

Since this inequality holds for all $\mathbf{y}\in\mathbb{R}^n$, it must also hold for the supremum over $\mathbf{y}$:

$$\sup_{\mathbf{y}\in\mathbb{R}^n} \left(\mathbf{x}^T\mathbf{y} - f^*(\mathbf{y})\right) \leq f(\mathbf{x}). \tag{19}$$

By definition of the biconjugate, this is exactly $f^{**}(\mathbf{x}) \leq f(\mathbf{x})$. □

Theorem 3.9 (Fenchel-Moreau): *Let $f : \mathbb{R}^n \to \mathbb{R}\cup\{+\infty\}$ be a proper function. Then $f = f^{**}$ if and only if f is **closed** (lower semi-continuous) and **convex**.*

Proof. The forward implication is immediate: since f^{**} is defined as the pointwise supremum of a family of affine functions, it is necessarily closed and convex. Thus, if $f = f^{**}$, f must share these properties.

For the reverse implication, assume f is proper, closed, and convex. By the weak duality lemma, $f^{**}(\mathbf{x}) \leq f(\mathbf{x})$ for all $\mathbf{x}\in\mathbb{R}^n$. It remains to show that $f(\mathbf{x}) \leq f^{**}(\mathbf{x})$.

We proceed by contradiction. Suppose there exists a point $\mathbf{x}_0 \in \mathbb{R}^n$ such that $f^{**}(\mathbf{x}_0) < f(\mathbf{x}_0)$. This implies that the point $(\mathbf{x}_0, f^{**}(\mathbf{x}_0)) \in \mathbb{R}^{n+1}$ lies strictly below the epigraph of f, denoted $\mathrm{epi}(f)$. Since f is closed and convex, $\mathrm{epi}(f)$ is a closed convex set.

By the **Strict Separating Hyperplane Theorem**, there exists a non-zero vector $(\mathbf{y}, \mu) \in \mathbb{R}^n \times \mathbb{R}$ and a scalar $\alpha \in \mathbb{R}$ such that:

$$\mathbf{y}^T\mathbf{x}_0 + \mu f^{**}(\mathbf{x}_0) > \alpha > \mathbf{y}^T\mathbf{x} + \mu t, \quad \forall(\mathbf{x}, t) \in \mathrm{epi}(f). \tag{20}$$

Since $(\mathbf{x}, t) \in \mathrm{epi}(f)$ allows t to be arbitrarily large, we must have $\mu \leq 0$ (otherwise the RHS would be unbounded).

Case 1: Non-vertical separation ($\mu < 0$) Divide the inequality by $-\mu > 0$. Let $\mathbf{m} = \mathbf{y}/(-\mu)$ and $c = \alpha/(-\mu)$. The separation becomes:

$$\mathbf{m}^T\mathbf{x}_0 - f^{**}(\mathbf{x}_0) > c > \mathbf{m}^T\mathbf{x} - t, \quad \forall(\mathbf{x}, t) \in \mathrm{epi}(f). \tag{21}$$

Setting $t = f(\mathbf{x})$ for $\mathbf{x} \in \mathrm{dom}(f)$, we obtain $\mathbf{m}^T\mathbf{x} - f(\mathbf{x}) < c$. Taking the supremum over $\mathbf{x}$:

$$f^*(\mathbf{m}) = \sup_{\mathbf{x}}(\mathbf{m}^T\mathbf{x} - f(\mathbf{x})) \leq c. \tag{22}$$

However, the left side of the separation inequality yields $\mathbf{m}^T\mathbf{x}_0 - c > f^{**}(\mathbf{x}_0)$. This implies that the affine function $h(\mathbf{z}) = \mathbf{m}^T\mathbf{z} - f^*(\mathbf{m})$ satisfies:

$$h(\mathbf{x}_0) = \mathbf{m}^T\mathbf{x}_0 - f^*(\mathbf{m}) \geq \mathbf{m}^T\mathbf{x}_0 - c > f^{**}(\mathbf{x}_0). \tag{23}$$

This contradicts the definition of f^{**} as the supremum of *all* such affine minorants: $f^{**}(\mathbf{x}_0) \geq h(\mathbf{x}_0)$.

Case 2: Vertical separation ($\mu = 0$) If $\mu = 0$, the separation reduces to $\mathbf{y}^T\mathbf{x}_0 > \alpha > \mathbf{y}^T\mathbf{x}$ for all $\mathbf{x} \in \mathrm{dom}(f)$. This means $\mathbf{x}_0$ is strictly separated from the domain of f in $\mathbb{R}^n$. Since f is proper and convex, it admits at least one affine minorant $g(\mathbf{z}) = \mathbf{a}^T\mathbf{z} - b \leq f(\mathbf{z})$. Consider the function $g_k(\mathbf{z}) = g(\mathbf{z}) + k(\mathbf{y}^T\mathbf{z} - \alpha)$. For $\mathbf{x} \in \mathrm{dom}(f)$, since $\mathbf{y}^T\mathbf{x} < \alpha$, the term $k(\mathbf{y}^T\mathbf{x} - \alpha)$ is negative. Thus, $g_k(\mathbf{x}) \leq g(\mathbf{x}) \leq f(\mathbf{x})$ for all $k > 0$. This means g_k is an affine minorant of f for any k. At $\mathbf{x}_0$, however, $\mathbf{y}^T\mathbf{x}_0 > \alpha$, so the term $k(\mathbf{y}^T\mathbf{x}_0 - \alpha) \to +\infty$ as $k \to \infty$. Consequently, $f^{**}(\mathbf{x}_0) \geq g_k(\mathbf{x}_0) \to +\infty$. Since we assumed $f^{**}(\mathbf{x}_0) < f(\mathbf{x}_0) < \infty$ (if $f(\mathbf{x}_0) = \infty$ the inequality holds trivially), this yields a contradiction.

Thus, the assumption $f^{**}(\mathbf{x}_0) < f(\mathbf{x}_0)$ is false. We conclude $f = f^{**}$. $\square$

Remark 3.10: *If f is convex but not closed, f^{**} is the lower semi-continuous closure of f. If f is not convex, f^{**} is the convex envelope of f.*

4 Duality

Corollary 4.1 (Fenchel-Young-Corollary): *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex and differentiable function. For any $\mathbf{x}, \mathbf{m} \in \mathbb{R}^n$, the following statements are equivalent:*

1. $\mathbf{m} = \nabla f(\mathbf{x})$
2. $f(\mathbf{x}) + f^*(\mathbf{m}) = \mathbf{m}^T\mathbf{x}$

Proof. Recall the Fenchel-Young inequality, which states $f(\mathbf{x}) + f^*(\mathbf{m}) \geq \mathbf{m}^T\mathbf{x}$ for all $\mathbf{x}, \mathbf{m}$. We prove the equivalence of the equality condition and the gradient condition.

(1 $\implies$ 2): Assume $\mathbf{m} = \nabla f(\mathbf{x})$. Since f is convex and differentiable, the first-order condition for convexity states that for all $\mathbf{y} \in \mathbb{R}^n$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}). \tag{24}$$

Substituting $\mathbf{m} = \nabla f(\mathbf{x})$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{m}^T\mathbf{y} - \mathbf{m}^T\mathbf{x}. \tag{25}$$

Rearranging terms to isolate $\mathbf{m}^T\mathbf{y} - f(\mathbf{y})$,

$$\mathbf{m}^T\mathbf{y} - f(\mathbf{y}) \leq \mathbf{m}^T\mathbf{x} - f(\mathbf{x}). \tag{26}$$

Taking the supremum of the left-hand side over all $\mathbf{y} \in \mathbb{R}^n$ yields the definition of the conjugate $f^*(\mathbf{m})$,

$$f^*(\mathbf{m}) = \sup_{\mathbf{y}}(\mathbf{m}^T\mathbf{y} - f(\mathbf{y})) \leq \mathbf{m}^T\mathbf{x} - f(\mathbf{x}). \tag{27}$$

This implies $f(\mathbf{x}) + f^*(\mathbf{m}) \leq \mathbf{m}^T\mathbf{x}$. Combining this with the Fenchel-Young inequality (which gives the reverse inequality $\geq$), we obtain equality,

$$f(\mathbf{x}) + f^*(\mathbf{m}) = \mathbf{m}^T\mathbf{x}. \tag{28}$$

(2 $\implies$ 1): Assume $f(\mathbf{x}) + f^*(\mathbf{m}) = \mathbf{m}^T\mathbf{x}$. Rearranging for $f(\mathbf{x})$,

$$f(\mathbf{x}) = \mathbf{m}^T\mathbf{x} - f^*(\mathbf{m}). \tag{29}$$

By definition of the conjugate, $f^*(\mathbf{m}) \geq \mathbf{m}^T\mathbf{y} - f(\mathbf{y})$ for any $\mathbf{y} \in \mathbb{R}^n$. Substituting this inequality into the equation above,

$$f(\mathbf{x}) \leq \mathbf{m}^T\mathbf{x} - (\mathbf{m}^T\mathbf{y} - f(\mathbf{y})). \tag{30}$$

Rearranging terms to compare $f(\mathbf{y})$ and $f(\mathbf{x})$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{m}^T(\mathbf{y} - \mathbf{x}). \tag{31}$$

This inequality holds for all $\mathbf{y} \in \mathbb{R}^n$. This is precisely the defining inequality for $\mathbf{m}$ being a subgradient of f at $\mathbf{x}$. Since f is differentiable, the subgradient is unique and equal to the gradient. Thus,

$$\mathbf{m} = \nabla f(\mathbf{x}). \tag{32}$$

□

Remark 4.2: *Geometrically, this corollary asserts that the vector $\mathbf{m}$ defines a supporting hyperplane to the graph of f at $(\mathbf{x}, f(\mathbf{x}))$ if and only if Fenchel's inequality holds with equality.*

4.1 Partial Minimization & Relation To Conjugate

Let $f : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R} \cup \{+\infty\}$ be a function of two variables $(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^k$.

Definition 4.3 (Partial Minimization): *The partial minimization of f with respect to $\mathbf{x}$ is the function $h : \mathbb{R}^k \to \mathbb{R} \cup \{+\infty\}$ defined as,*

$$h(\mathbf{y}) = \inf_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}, \mathbf{y}). \tag{33}$$

Lemma 4.4 (Preservation of Convexity): *If f is jointly convex in $(\mathbf{x}, \mathbf{y})$, then $h(\mathbf{y})$ is a convex function.*

Remark 4.5:

$$f^*(\mathbf{m}_1, \mathbf{m}_2) = \sup_{\mathbf{x},\mathbf{y}} \left(\mathbf{m}_1^T \mathbf{x} + \mathbf{m}_2^T \mathbf{y} - f(\mathbf{x}, \mathbf{y})\right) \tag{34}$$

Remark 4.6: $h^*(\mathbf{m}) = f^*(0, \mathbf{m})$

Proof.

$$\begin{aligned} h^*(\mathbf{m}) &= \sup_{\mathbf{y}} \left(\mathbf{m}^T \mathbf{y} - h(\mathbf{y})\right) \\ &= \sup_{\mathbf{y}} \left(\mathbf{m}^T \mathbf{y} - \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y})\right) \\ &= \sup_{\mathbf{y}} \sup_{\mathbf{x}} \left(\mathbf{m}^T \mathbf{y} - f(\mathbf{x}, \mathbf{y})\right) = f^*(0, \mathbf{m}) \end{aligned} \tag{35}$$

□

Remark 4.7: $\sup\left[-f^*(0, \mathbf{m})\right] \leq \inf f(\mathbf{x}, 0)$ *i.e.* $h^{**}(0) \leq h(0)$

Remark 4.8: *Partial minimization appears naturally when eliminating primal variables in dual constructions.*

4.1.1 Perturbation Interpretation Of Duality

For any problem $p^* = \min_{\mathbf{x}} f(\mathbf{x})$, "the" dual is constructed by

1. Introduce perturbations $\mathbf{y}$, and extend f such that $f(\mathbf{x}, 0)$ returns the function to minimize
2. the dual is $d^* = \sup\left[-f^*(0, \mathbf{m})\right]$

   where $\mathbf{m}$ is the dual variable and $\mathbf{x}$ is the primal variable.
   We have

$$d^* \leq p^*, \tag{36}$$

under convexity + mild conditions, we have strong duality, i.e. $d^* = p^*$.

5 Lagrangian Duality

Given the optimization problem in standard form:

$$p^* = \inf\{f_0(x) \mid f_i(x) \leq 0,\ i = 1, \ldots, m,\ x \in \mathcal{D}\} \tag{37}$$

where $\mathcal{D} \subseteq \mathbb{R}^n$ is the intersection of the domains of all functions f_i.

5.1 The Lagrangian and Dual Function

Define the **Lagrangian** $L : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ for $\lambda \succeq 0$ as,

$$L(x, \lambda) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x). \tag{38}$$

The **Lagrange dual function** $g : \mathbb{R}^m \to \mathbb{R}$ is defined as the pointwise infimum of the Lagrangian over x,

$$g(\lambda) = \inf_{x \in \mathcal{D}} L(x, \lambda). \tag{39}$$

By the property of pointwise infima, g is a concave function, and it provides a lower bound on the primal optimal value: $g(\lambda) \leq p^*$ for all $\lambda \succeq 0$.

The Dual Problem

The **Lagrange dual problem** seeks the tightest lower bound,

$$\begin{aligned} d^* = \sup_{\lambda} \quad & g(\lambda) \\ \text{s.t.} \quad & \lambda \succeq 0 \end{aligned} \tag{40}$$

The relationship between the primal and dual optimal values is governed by **Weak Duality** ($d^* \leq p^*$). Under constraint qualifications (e.g., Slater's condition) and convexity, **Strong Duality** holds ($d^* = p^*$).

6 Shadow Price Interpretation of Duality

7 Game Theoretic Interpretation of Duality

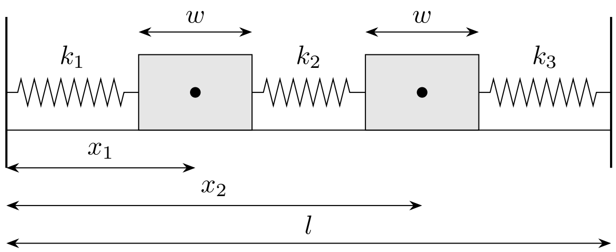
8 KKT Conditions

Consider a standard optimization problem with strong duality and differentiable constraints. Let x^* be the primal optima and λ^* be the dual optima.

Definition 8.1 (KKT Conditions): *The following statements constitute the KKT conditions,*

1. $f_i(x^*) \leq 0$ *&* $h_j(x^*) = 0$, $\forall i, j$ *(Primal feasibility).*
2. $\lambda_i^* \geq 0$, $\forall i$ *(Dual feasibility).*
3. $\lambda_i^* f_i(x^*) = 0$, $\forall i$ *(Complementary slackness).*
4. $\nabla_x L(x^*, \lambda) = 0$.

9 Mechanistic Interpretation of KKT Conditions



There's maybe no need for this.
Yeah, there's no need for this.

10 Multicriterion Optimization

Some shit. Write it.

11 Moreau Envelope (Continued)

Continued from here.

Example

$$p^* = \min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2 \qquad g(x) = p^* \tag{41}$$

Reframe as:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{y}} \quad & \|\mathbf{y}\|_2 \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} - \mathbf{b} = \mathbf{y} \end{aligned} \tag{42}$$

Lagrangian:

$$L(\mathbf{x}, \mathbf{y}, \mathbf{v}) = \|\mathbf{y}\|_2 + \mathbf{v}^T(\mathbf{A}\mathbf{x} - \mathbf{b} - \mathbf{y}) \tag{43}$$

Dual Function:

$$\begin{aligned} g(\mathbf{v}) &= \inf_{\mathbf{x},\mathbf{y}} \left[\|\mathbf{y}\|_2 + \mathbf{v}^T(\mathbf{A}\mathbf{x} - \mathbf{b} - \mathbf{y})\right] \\ &= -\mathbf{v}^T\mathbf{b} + \left[\inf_{\mathbf{x},\mathbf{y}} \left(\underbrace{\|\mathbf{y}\|_2 - \mathbf{v}^T\mathbf{y}}_{\substack{-\infty \text{ if } \|\mathbf{v}\|_2 > 1 \\ 0 \text{ otherwise}}}\right) + \left(\underbrace{(\mathbf{A}^T\mathbf{v})^T\mathbf{x}}_{\substack{0 \text{ if } \mathbf{A}^T\mathbf{v} = 0 \\ -\infty \text{ otherwise}}}\right)\right] \end{aligned} \tag{44}$$

Result:

$$g(\mathbf{v}) = \begin{cases} -\mathbf{v}^T\mathbf{b} & \text{if } \mathbf{A}^T\mathbf{v} = 0 \;\&\; \|\mathbf{v}\|_2 \leq 1 \\ -\infty & \text{otherwise} \end{cases} \tag{45}$$

11.1 Dual of Moreau envelope

Let $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be a proper, closed, convex function. Recall the Moreau envelope $f_u(x)$ with parameter $u > 0$ is defined as the infimal convolution of f and a quadratic kernel,

$$f_u(x) = \inf_y \left\{ f(y) + \frac{1}{2u}\|x - y\|^2 \right\}. \tag{46}$$

To derive its dual representation, we rewrite this as a constrained minimization problem by introducing an auxiliary variable z

$$\begin{aligned} \min_{y,z} \quad & f(y) + \frac{1}{2u}\|z\|^2 \\ \text{s.t.} \quad & x - y = z. \end{aligned} \tag{47}$$

The dual function $g(v)$ is given by

$$g(v) = \langle v, x\rangle - f^*(v) - \frac{u}{2}\|v\|^2, \tag{48}$$

which leads to the identity

$$f_u(x) = \left(f^* + \frac{u}{2}\|\cdot\|^2\right)^*(x). \tag{49}$$

Proof. We form the Lagrangian $L(y, z, v)$ with dual variable $v \in \mathbb{R}^n$,

$$L(y, z, v) = f(y) + \frac{1}{2u}\|z\|^2 + \langle v, x - y - z\rangle. \tag{50}$$

The Lagrange dual function $g(v)$ is obtained by minimizing the Lagrangian with respect to the primal variables y and z,

$$\begin{aligned} g(v) &= \inf_{y,z} L(y, z, v) \\ &= \inf_{y,z} \left\{ f(y) + \frac{1}{2u}\|z\|^2 + \langle v, x\rangle - \langle v, y\rangle - \langle v, z\rangle \right\} \\ &= \langle v, x\rangle + \inf_y \left\{ f(y) - \langle v, y\rangle \right\} + \inf_z \left\{ \frac{1}{2u}\|z\|^2 - \langle v, z\rangle \right\}. \end{aligned} \tag{51}$$

We evaluate the infima term-wise:

1. Term involving y: Using the definition of the Fenchel conjugate, $f^*(v) = \sup_y\{\langle v, y\rangle - f(y)\}$, we have,

$$\inf_y\{f(y) - \langle v, y\rangle\} = -\sup_y\{\langle v, y\rangle - f(y)\} = -f^*(v). \tag{52}$$

2. Term involving z: This is the conjugate of the quadratic function $h(z) = \frac{1}{2u}\|z\|^2$. Setting the gradient with respect to z to zero,

$$\nabla_z \left(\frac{1}{2u}\|z\|^2 - \langle v, z\rangle \right) = \frac{z}{u} - v = 0 \implies z = uv. \tag{53}$$

Substituting $z = uv$ back into the expression,

$$\frac{1}{2u}\|uv\|^2 - \langle v, uv\rangle = \frac{u}{2}\|v\|^2 - u\|v\|^2 = -\frac{u}{2}\|v\|^2. \tag{54}$$

Combining these, the dual function

$$g(v) = \langle v, x\rangle - f^*(v) - \frac{u}{2}\|v\|^2. \tag{55}$$

The dual problem consists of maximizing $g(v)$. Since f is proper, closed, and convex, strong duality holds ($p^* = d^*$), and we can express the primal optimum as the supremum of the dual:

$$\begin{aligned} f_u(x) &= \sup_v g(v) \\ &= \sup_v \left\{ \langle v, x\rangle - \left(f^*(v) + \frac{u}{2}\|v\|^2 \right) \right\}. \end{aligned} \tag{56}$$

Recognizing the right-hand side as the Fenchel conjugate of the function $\phi(v) = f^*(v) + \frac{u}{2}\|v\|^2$ evaluated at x, we conclude:

$$f_u(x) = \left(f^* + \frac{u}{2}\|\cdot\|^2 \right)^* (x). \tag{57}$$

□

11.2 Moreau Decomposition

12 Proximal Algorithms

12.1 Fixed Point

13 Tutorial 1: Projection on ℓ_1 ball

14 Statistical Interpretation of Norm Approximation

15 Analysis of Least Squares

15.1 Problem Setup and Definitions

Let us define a dataset comprising a fixed design matrix $\mathbf{A} \in \mathbb{R}^{n\times d}$ (where $n > d$) representing the feature vectors, and a random response vector $\mathbf{y} \in \mathbb{R}^n$. We assume the labels are generated according to a linear ground-truth model:

$$\mathbf{y} = \mathbf{A}\mathbf{x}^* + \boldsymbol{\epsilon} \tag{58}$$

where $\mathbf{x}^* \in \mathbb{R}^d$ is the true, unknown parameter vector, and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_n)$ is the observation noise vector. The goal is to learn an estimator $\hat{\mathbf{x}}$ that closely approximates $\mathbf{y} \approx \mathbf{A}\hat{\mathbf{x}}$.

Definition 15.1 (Risk and Excess Risk): *Let $\tilde{\mathbf{y}} = \mathbf{A}\mathbf{x}^* + \tilde{\boldsymbol{\epsilon}}$ be an independent test sample drawn from the identical data-generating process, where $\tilde{\boldsymbol{\epsilon}} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ is independent of $\boldsymbol{\epsilon}$. The **Risk** of an estimator $\hat{\mathbf{x}}$ is defined as:*

$$\mathcal{R}(\hat{\mathbf{x}}) = \frac{1}{n}\mathbb{E}_{\boldsymbol{\epsilon},\tilde{\boldsymbol{\epsilon}}} \|\tilde{\mathbf{y}} - \mathbf{A}\hat{\mathbf{x}}\|^2 \tag{59}$$

*The **Excess Risk** isolates the error attributable to the estimation of $\mathbf{x}^*$ by removing the irreducible noise σ^2:*

$$\mathcal{E}(\hat{\mathbf{x}}) = \frac{1}{n}\mathbb{E}_{\boldsymbol{\epsilon}} \|\mathbf{A}(\mathbf{x}^* - \hat{\mathbf{x}})\|^2 \tag{60}$$

Lemma 15.2 (Bias-Variance Decomposition): *The excess risk of any estimator $\hat{\mathbf{x}}$ can be additively decomposed into an approximation error (Bias term) and a prediction error (Variance term):*

$$\mathcal{E}(\hat{\mathbf{x}}) = \underbrace{\frac{1}{n}\mathbb{E}_{\boldsymbol{\epsilon}} \|\mathbf{A}(\mathbf{x}^* - \mathbb{E}[\hat{\mathbf{x}}])\|^2}_{Bias} + \underbrace{\frac{1}{n}\mathbb{E}_{\boldsymbol{\epsilon}} \|\mathbf{A}(\hat{\mathbf{x}} - \mathbb{E}[\hat{\mathbf{x}}])\|^2}_{Variance} \tag{61}$$

Proof. By adding and subtracting $\mathbb{E}[\hat{\mathbf{x}}]$, we have $\mathbf{A}(\mathbf{x}^* - \hat{\mathbf{x}}) = \mathbf{A}(\mathbf{x}^* - \mathbb{E}[\hat{\mathbf{x}}]) + \mathbf{A}(\mathbb{E}[\hat{\mathbf{x}}] - \hat{\mathbf{x}})$. Taking the expected squared norm, the cross-term $\frac{2}{n}\mathbb{E}[(\mathbf{A}(\mathbf{x}^* - \mathbb{E}[\hat{\mathbf{x}}]))^\top \mathbf{A}(\mathbb{E}[\hat{\mathbf{x}}] - \hat{\mathbf{x}})]$ vanishes because $\mathbb{E}[\mathbb{E}[\hat{\mathbf{x}}] - \hat{\mathbf{x}}] = \mathbf{0}$. The remaining terms form the bias and variance components. □

15.2 Analysis of Ordinary Least Squares (OLS)

Theorem 15.3 (Risk of OLS): *The Ordinary Least Squares estimator, defined as $\hat{\mathbf{x}}_{OLS} = \arg\min_{\mathbf{x}} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2$, is an unbiased estimator with an excess risk of exactly $\frac{\sigma^2 d}{n}$.*

Proof. The closed-form solution is $\hat{\mathbf{x}}_{\mathrm{OLS}} = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{y}$. Taking the expectation:

$$\mathbb{E}[\hat{\mathbf{x}}_{\mathrm{OLS}}] = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbb{E}[\mathbf{A}\mathbf{x}^* + \boldsymbol{\epsilon}] = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{A}\mathbf{x}^* = \mathbf{x}^* \tag{62}$$

Since the estimator is unbiased, the excess risk is entirely variance. As derived in standard regression analysis, the variance evaluates to $\frac{\sigma^2 d}{n}$. □

15.3 Analysis of Ridge Regression

To mitigate the variance term characteristic of OLS, particularly in settings where d is large relative to n, we introduce ℓ_2 regularization.

Definition 15.4 (Ridge Estimator): *For a regularization parameter $\lambda > 0$, the Ridge estimator is:*

$$\hat{\mathbf{x}}_{ridge} = \arg\min_{\mathbf{x}} \left[\frac{1}{n}\|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2 + \lambda\|\mathbf{x}\|^2\right] = \frac{1}{n}(\boldsymbol{\Sigma} + \lambda\mathbf{I})^{-1}\mathbf{A}^\top\mathbf{y} \tag{63}$$

where $\boldsymbol{\Sigma} = \frac{1}{n}\mathbf{A}^\top\mathbf{A}$ is the empirical covariance matrix.

By substituting $\mathbf{y} = \mathbf{A}\mathbf{x}^* + \boldsymbol{\epsilon}$, the expected Ridge estimator is $\mathbb{E}[\hat{\mathbf{x}}_{\mathrm{ridge}}] = (\boldsymbol{\Sigma} + \lambda\mathbf{I})^{-1}\boldsymbol{\Sigma}\mathbf{x}^*$, demonstrating that Ridge regression introduces bias.

Lemma 15.5 (Ridge Bias Bound): *The bias term for the Ridge estimator is bounded by $\frac{\lambda}{4}\|\mathbf{x}^*\|^2$.*

Proof. Define $\Delta \mathbf{x} \triangleq \mathbf{x}^* - \mathbb{E}[\hat{\mathbf{x}}_{\text{ridge}}]$. Then:

$$\Delta \mathbf{x} = \left(\mathbf{I} - (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{\Sigma}\right) \mathbf{x}^* = \lambda (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{x}^*. \tag{64}$$

The bias evaluates to:

$$\text{Bias}(\hat{\mathbf{x}}_{\text{ridge}}) \triangleq \frac{1}{n} \|\mathbf{A} \Delta \mathbf{x}\|^2 = (\Delta \mathbf{x})^\top \mathbf{\Sigma} (\Delta \mathbf{x}). \tag{65}$$

Substituting $\Delta \mathbf{x}$:

$$\text{Bias}(\hat{\mathbf{x}}_{\text{ridge}}) = \lambda^2 (\mathbf{x}^*)^\top (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{\Sigma} (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{x}^*. \tag{66}$$

Since $[\mathbf{\Sigma}, (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1}] = \mathbf{0}$:

$$\text{Bias}(\hat{\mathbf{x}}_{\text{ridge}}) = (\mathbf{x}^*)^\top \underbrace{\left[\lambda^2 \mathbf{\Sigma} (\mathbf{\Sigma} + \lambda \mathbf{I})^{-2}\right]}_{\triangleq \mathbf{M}} \mathbf{x}^*. \tag{67}$$

Let $\mathbf{\Sigma} = \mathbf{V} \mathbf{D} \mathbf{V}^\top$ denote the spectral decomposition, where $\mathbf{D} = \text{diag}(\mu_1, \ldots, \mu_d)$ and $\mu_i \geq 0 \; \forall i$. The eigenvalues of $\mathbf{M}$ are given by $\lambda_i(\mathbf{M}) = \frac{\lambda^2 \mu_i}{(\mu_i + \lambda)^2}$.

Applying the inequality $(\mu_i + \lambda)^2 = (\mu_i - \lambda)^2 + 4\mu_i \lambda \geq 4 \mu_i \lambda$:

$$\lambda_{\max}(\mathbf{M}) = \max_i \frac{\lambda^2 \mu_i}{(\mu_i + \lambda)^2} \leq \frac{\lambda^2 \mu_i}{4 \mu_i \lambda} = \frac{\lambda}{4}. \tag{68}$$

By the Rayleigh-Ritz theorem, $\sup_{\mathbf{x}^* \neq \mathbf{0}} \frac{(\mathbf{x}^*)^\top \mathbf{M} \mathbf{x}^*}{\|\mathbf{x}^*\|^2} = \lambda_{\max}(\mathbf{M})$. Consequently:

$$\text{Bias}(\hat{\mathbf{x}}_{\text{ridge}}) \leq \lambda_{\max}(\mathbf{M}) \|\mathbf{x}^*\|^2 \leq \frac{\lambda}{4} \|\mathbf{x}^*\|^2. \tag{69}$$

□

Lemma 15.6 (Ridge Variance Bound): *The variance term for the Ridge estimator is bounded by* $\frac{\sigma^2 \operatorname{Tr}(\mathbf{\Sigma})}{4n\lambda}$.

Proof. Define the estimator deviation $\delta \hat{\mathbf{x}} \triangleq \hat{\mathbf{x}}_{\text{ridge}} - \mathbb{E}[\hat{\mathbf{x}}_{\text{ridge}}] = \frac{1}{n}(\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \boldsymbol{\epsilon}$. Let the prediction variance be denoted by $\text{Var}(\hat{\mathbf{x}}_{\text{ridge}}) \triangleq \frac{1}{n} \mathbb{E}\left[\|\mathbf{A} \delta \hat{\mathbf{x}}\|^2\right] = \mathbb{E}\left[\|\mathbf{C} \boldsymbol{\epsilon}\|^2\right]$, where $\mathbf{C} \triangleq \frac{1}{n\sqrt{n}} \mathbf{A} (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top$.

Assuming $\boldsymbol{\epsilon} \sim (\mathbf{0}, \sigma^2 \mathbf{I})$, the expectation of the quadratic form simplifies to:

$$\text{Var}(\hat{\mathbf{x}}_{\text{ridge}}) = \sigma^2 \operatorname{Tr}(\mathbf{C}^\top \mathbf{C}) = \frac{\sigma^2}{n^3} \operatorname{Tr}\left(\mathbf{A} (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{A} (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top\right). \tag{70}$$

Using the cyclic property of the trace and substituting $\mathbf{\Sigma} = \frac{1}{n} \mathbf{A}^\top \mathbf{A}$:

$$\begin{aligned} \text{Var}(\hat{\mathbf{x}}_{\text{ridge}}) &= \frac{\sigma^2}{n} \operatorname{Tr}\left(\left(\frac{1}{n} \mathbf{A}^\top \mathbf{A}\right) (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \left(\frac{1}{n} \mathbf{A}^\top \mathbf{A}\right) (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1}\right) \\ &= \frac{\sigma^2}{n} \operatorname{Tr}\left(\mathbf{\Sigma}^2 (\mathbf{\Sigma} + \lambda \mathbf{I})^{-2}\right). \end{aligned} \tag{71}$$

Let $\text{spec}(\mathbf{\Sigma}) = \{\mu_i\}_{i=1}^d$. Expressing the trace in terms of the spectrum gives:

$$\operatorname{Tr}\left(\mathbf{\Sigma}^2 (\mathbf{\Sigma} + \lambda \mathbf{I})^{-2}\right) = \sum_i \frac{\mu_i^2}{(\mu_i + \lambda)^2}. \tag{72}$$

Applying the previously established inequality $\frac{\mu_i}{(\mu_i+\lambda)^2} \le \frac{1}{4\lambda}$:

$$\sum_i \frac{\mu_i^2}{(\mu_i + \lambda)^2} = \sum_i \mu_i \left(\frac{\mu_i}{(\mu_i + \lambda)^2} \right) \le \frac{1}{4\lambda} \sum_i \mu_i = \frac{\mathrm{Tr}(\boldsymbol{\Sigma})}{4\lambda}. \tag{73}$$

Consequently:

$$\mathrm{Var}(\hat{\mathbf{x}}_{\mathrm{ridge}}) \le \frac{\sigma^2 \,\mathrm{Tr}(\boldsymbol{\Sigma})}{4n\lambda}. \tag{74}$$

□

Theorem 15.7 (Excess Risk of Ridge Regression): *By selecting the optimal regularization parameter $\lambda = \frac{\sigma\sqrt{\mathrm{Tr}(\boldsymbol{\Sigma})}}{\|\mathbf{x}^*\|\sqrt{n}}$, the excess risk of the Ridge estimator is bounded by $\mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$.*

Proof. Let $\mathcal{E}(\hat{\mathbf{x}}_{\mathrm{ridge}})$ denote the excess risk. Combining the preceding bounds on bias and variance yields:

$$\mathcal{E}(\hat{\mathbf{x}}_{\mathrm{ridge}}) \le \underbrace{\frac{\|\mathbf{x}^*\|^2}{4}}_{\triangleq a} \lambda + \underbrace{\frac{\sigma^2 \,\mathrm{Tr}(\boldsymbol{\Sigma})}{4n}}_{\triangleq b} \frac{1}{\lambda}. \tag{75}$$

Define the upper bound function $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $f(\lambda) \triangleq a\lambda + b\lambda^{-1}$.

Applying the Arithmetic Mean-Geometric Mean (AM-GM) inequality:

$$f(\lambda) \ge 2\sqrt{(a\lambda)(b\lambda^{-1})} = 2\sqrt{ab}. \tag{76}$$

The minimum is uniquely attained when $a\lambda = b\lambda^{-1}$, yielding the optimal regularization parameter $\lambda^* = \sqrt{\frac{b}{a}}$.

Evaluating the infimum of the bound gives:

$$\inf_{\lambda>0} f(\lambda) = 2\sqrt{ab} = 2\sqrt{\left(\frac{\|\mathbf{x}^*\|^2}{4}\right)\left(\frac{\sigma^2 \,\mathrm{Tr}(\boldsymbol{\Sigma})}{4n}\right)} = \frac{\sigma \|\mathbf{x}^*\| \sqrt{\mathrm{Tr}(\boldsymbol{\Sigma})}}{2\sqrt{n}}. \tag{77}$$

Consequently, the minimal excess risk bound is given by:

$$\inf_{\lambda>0} \mathcal{E}(\hat{\mathbf{x}}_{\mathrm{ridge}}) \le \frac{\sigma \|\mathbf{x}^*\| \sqrt{\mathrm{Tr}(\boldsymbol{\Sigma})}}{2\sqrt{n}}. \tag{78}$$

□

16 Gradient Descent

Let $f : \mathbb{R}^d \to \mathbb{R}$ be a continuously differentiable objective function. We consider the unconstrained optimization problem $\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x})$. The standard Gradient Descent (GD) algorithm generates a sequence of iterates $\{\mathbf{x}^{(k)}\}_{k=0}^{\infty}$ via the recurrence relation

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \eta \nabla f(\mathbf{x}^{(k)}) \tag{79}$$

where $\eta > 0$ is the step size. To rigorously guarantee convergence, we must impose structural conditions on the objective function f.

Definition 16.1 (L-Smoothness and Strong Convexity): *A differentiable function f is* ***L-smooth*** *if its gradient is Lipschitz continuous with constant $L > 0$:*

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2 \leq L \|\mathbf{x} - \mathbf{y}\|_2 \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d \tag{80}$$

Furthermore, f is ***μ-strongly convex*** *for $\mu > 0$ if:*

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d \tag{81}$$

If f is twice continuously differentiable, these conditions are equivalent to bounding the spectrum of the Hessian matrix: $\mu \mathbf{I} \preceq \nabla^2 f(\mathbf{x}) \preceq L\mathbf{I}$ for all $\mathbf{x} \in \mathbb{R}^d$, where $\preceq$ denotes the Loewner partial order.

16.1 The Least Squares Objective

Let $\mathbf{A} \in \mathbb{R}^{n \times d}$ ($n \geq d$) denote a fixed design matrix and $\mathbf{y} \in \mathbb{R}^n$ a target vector. The normalized ordinary least squares (OLS) problem seeks to minimize the empirical risk:

$$f(\mathbf{x}) = \frac{1}{2n} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 \tag{82}$$

Lemma 16.2 (Properties of the OLS Objective): *Assume $\mathbf{A}$ has full column rank (rank($\mathbf{A}$) $= d$). Let $\mathbf{H} \triangleq \frac{1}{n}\mathbf{A}^\top\mathbf{A}$ be the Hessian matrix. The objective $f(\mathbf{x})$ is L-smooth and μ-strongly convex with parameters:*

$$L = \lambda_{\max}(\mathbf{H}), \quad \mu = \lambda_{\min}(\mathbf{H}) \tag{83}$$

where $\lambda_{\max}$ and $\lambda_{\min}$ denote the maximal and minimal eigenvalues of $\mathbf{H}$, respectively. Moreover, the unique global minimizer is $\mathbf{x}^* = \mathbf{H}^{-1}\left(\frac{1}{n}\mathbf{A}^\top\mathbf{y}\right)$.

Proof. The gradient is explicitly given by $\nabla f(\mathbf{x}) = \mathbf{H}\mathbf{x} - \frac{1}{n}\mathbf{A}^\top\mathbf{y}$. The Hessian is the constant symmetric matrix $\nabla^2 f(\mathbf{x}) = \mathbf{H}$. Since $\mathbf{A}$ has full column rank, $\mathbf{A}^\top\mathbf{A}$ is positive definite, implying $\mu = \lambda_{\min}(\mathbf{H}) > 0$. Thus, $\mu\mathbf{I} \preceq \mathbf{H} \preceq L\mathbf{I}$, fulfilling Definition L-Smoothness and Strong Convexity globally. Setting $\nabla f(\mathbf{x}^*) = \mathbf{0}$ directly yields the unique optimum due to the invertibility of $\mathbf{H}$. □

16.2 Spectral Error Dynamics

We analyze the evolution of the error vector $\mathbf{e}^{(k)} \triangleq \mathbf{x}^{(k)} - \mathbf{x}^*$.

Theorem 16.3 (Exact Error Operator): *For the OLS objective, the error sequence under Gradient Descent evolves according to the linear operator $(\mathbf{I} - \eta\mathbf{H})$:*

$$\mathbf{e}^{(k)} = (\mathbf{I} - \eta\mathbf{H})^k \mathbf{e}^{(0)} \tag{84}$$

Proof. By substituting the gradient into the GD update rule:

$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \eta\left(\mathbf{H}\mathbf{x}^{(k-1)} - \frac{1}{n}\mathbf{A}^\top\mathbf{y}\right) \tag{85}$$

Utilizing the optimality condition $\frac{1}{n}\mathbf{A}^\top \mathbf{y} = \mathbf{H}\mathbf{x}^*$, we have:

$$\begin{aligned} \mathbf{x}^{(k)} &= \mathbf{x}^{(k-1)} - \eta\mathbf{H}(\mathbf{x}^{(k-1)} - \mathbf{x}^*) \\ \mathbf{x}^{(k)} - \mathbf{x}^* &= (\mathbf{I} - \eta\mathbf{H})(\mathbf{x}^{(k-1)} - \mathbf{x}^*) \end{aligned} \tag{86}$$

Applying this relation recursively k times yields the stated theorem. □

To bound $\left\|\mathbf{e}^{(k)}\right\|_2$, we must constrain the spectral radius $\rho(\mathbf{I} - \eta\mathbf{H})$. By the Spectral Theorem, the symmetric positive definite matrix $\mathbf{H}$ admits an orthogonal eigendecomposition $\mathbf{H} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$, where $\mathbf{V}$ is an orthogonal matrix and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_d)$ contains the eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d > 0$. [1] fdsf

Theorem 16.4: *For any real symmetric matrix $\mathbf{M} \in \mathbb{R}^{d\times d}$ and vector $v \in \mathbb{R}^d$,*

$$\lambda_{\min} v^T v \leq v^T M v \leq \lambda_{\max} v^T v \tag{87}$$

where $\lambda_{\min}$ and $\lambda_{\max}$ are the minimum and maximum eigenvalues of M.

Corollary 16.5 (Spectral Bound and Convergence): *The induced ℓ_2-norm of the iteration operator is bounded by:*

$$\|\mathbf{I} - \eta\mathbf{H}\|_2 = \max_{1\leq i\leq d} |1 - \eta\lambda_i| \tag{88}$$

Consequently, $\lim_{k\to\infty} \mathbf{x}^{(k)} = \mathbf{x}^*$ for any initialization $\mathbf{x}^{(0)}$ if and only if $0 < \eta < \frac{2}{L}$.

16.3 Optimal Step Size via Minimax Formulation

We define the condition number of the problem as $\kappa \triangleq \frac{L}{\mu} = \frac{\lambda_1}{\lambda_d} \geq 1$. The contraction rate is defined as $q(\eta) \triangleq \max_i |1 - \eta\lambda_i|$. To maximize the convergence speed, we seek the step size η^* that minimizes this worst-case spectral radius.

Theorem 16.6 (Optimal Constant Step Size): *The optimal fixed step size minimizing the spectral radius is:*

$$\eta^* = \frac{2}{L + \mu} \tag{89}$$

yielding an optimal contraction factor of $q(\eta^*) = \frac{\kappa-1}{\kappa+1}$.

Proof. We formulate the minimax optimization problem:

$$\eta^* = \arg\min_{\eta>0} \max_{1\leq i\leq d} |1 - \eta\lambda_i| \tag{90}$$

Because $1 - \eta\lambda$ is a strictly decreasing function of λ for any fixed $\eta > 0$, the maximum absolute value must occur at the boundaries of the spectrum: $\lambda_1 = L$ or $\lambda_d = \mu$. Thus:

$$\max_{1\leq i\leq d} |1 - \eta\lambda_i| = \max\left(|1 - \eta\mu|, |1 - \eta L|\right) \tag{91}$$

[1] Recall the standard linear algebra bounds for a real symmetric matrix $\mathbf{M}$ with eigenvalues λ_i: the quadratic form is bounded by the Rayleigh quotient $\lambda_{\min} \|\mathbf{v}\|_2^2 \leq \mathbf{v}^\top \mathbf{M}\mathbf{v} \leq \lambda_{\max} \|\mathbf{v}\|_2^2$. Furthermore, its induced ℓ_2-norm (spectral norm) equates to its spectral radius: $\|\mathbf{M}\|_2 = \max_i |\lambda_i|$.

For a valid learning rate ($\eta < 2/L$), the term $(1 - \eta\mu)$ is strictly positive and linearly decreasing, while $(1-\eta L)$ becomes negative, making $|1-\eta L| = \eta L - 1$ linearly increasing. The global minimum of the maximum of these two intersecting linear functions occurs precisely where they are equal:

$$1 - \eta\mu = \eta L - 1 \implies \eta(L+\mu) = 2 \implies \eta^* = \frac{2}{L+\mu} \tag{92}$$

Substituting η^* back into the contraction factor yields:

$$q(\eta^*) = 1 - \left(\frac{2}{L+\mu}\right)\mu = \frac{L-\mu}{L+\mu} = \frac{(L/\mu)-1}{(L/\mu)+1} = \frac{\kappa-1}{\kappa+1} \tag{93}$$

□

Remark 16.7 (Iteration Complexity): *Given the optimal rate $q(\eta^*) = \frac{\kappa-1}{\kappa+1} \approx 1 - \frac{2}{\kappa}$ (for large κ), the error bound is $\left\|\mathbf{e}^{(k)}\right\|_2 \leq \left(1-\frac{1}{\kappa}\right)^{2k} \left\|\mathbf{e}^{(0)}\right\|_2 \leq \exp\left(-\frac{2k}{\kappa}\right) \left\|\mathbf{e}^{(0)}\right\|_2$. To guarantee $\left\|\mathbf{e}^{(k)}\right\|_2 \leq \epsilon$, the required number of iterations scales as $\mathcal{O}\left(\kappa \log \frac{1}{\epsilon}\right)$.*

16.4 Computational Complexity: Exact Direct Methods

Finding the exact optimal solution $\mathbf{x}^*$ can be done via several matrix factorizations, all of which generally scale as $\mathcal{O}(nd^2)$ but differ in practical constants and numerical stability:

1. **Normal Equations (Cholesky):** $\mathbf{x}^* = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{y}$. Computing the Hessian $\mathbf{A}^\top\mathbf{A}$ takes $\mathcal{O}(nd^2)$, and its inversion/factorization takes $\mathcal{O}(d^3)$. Total time is $\mathcal{O}(nd^2)$.
2. **QR Decomposition:** $\mathbf{A} = \mathbf{QR}$. The solution is obtained by solving the triangular system $\mathbf{R}\mathbf{x}^* = \mathbf{Q}^\top\mathbf{y}$, taking $\mathcal{O}(nd^2)$.
3. **Singular Value Decomposition (SVD):** $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, implying $\mathbf{x}^* = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top\mathbf{y}$. This is the most numerically stable but computationally heaviest $\mathcal{O}(nd^2)$ approach.

In practice, execution times follow: Cholesky < QR < SVD.

16.5 Gradient Descent (GD) Iterations

GD is an approximate method requiring $k = \mathcal{O}\left(\kappa \log \frac{1}{\epsilon}\right)$ iterations to reach ϵ-accuracy, where κ is the condition number of the Hessian $\mathbf{H} = \frac{1}{n}\mathbf{A}^\top\mathbf{A}$. We have two primary ways to compute the gradient step $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \eta\nabla f(\mathbf{x}^{(k)})$:

Lemma 16.8 (GD Complexity Regimes): *Gradient Descent can be implemented in two distinct computational paradigms:*

- ***Approach 1 (Precomputation):*** $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \eta\left(\mathbf{H}\mathbf{x}^{(k)} - \frac{1}{n}\mathbf{A}^\top\mathbf{y}\right)$. *We precompute $\mathbf{H}$ and $\mathbf{A}^\top\mathbf{y}$ in $\mathcal{O}(nd^2)$. Each subsequent iteration takes $\mathcal{O}(d^2)$. Total complexity: $\mathcal{O}(nd^2 + d^2\kappa \log \frac{1}{\epsilon})$.*
- ***Approach 2 (No Precomputation):*** $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \frac{\eta}{n}\mathbf{A}^\top(\mathbf{A}\mathbf{x}^{(k)} - \mathbf{y})$. *We compute the matrix-vector products iteratively. Each iteration takes $\mathcal{O}(nd)$. Total complexity: $\mathcal{O}(nd\kappa \log \frac{1}{\epsilon})$.*

Remark 16.9 (Why use Gradient Descent?): *In regimes where d is very large, Approach 2 avoids the $\mathcal{O}(nd^2)$ cost entirely, offering significant computational savings if the problem is well-conditioned (κ is small). Furthermore, GD is favored for:*

1. ***Memory Restrictions:*** *Iterative methods avoid storing the $d \times d$ dense Hessian, making them suitable for streaming algorithms or massive datasets.*

2. ***Implicit Regularization:*** *When stopped early, GD provides better generalization error by preventing overfitting to the noise, acting akin to Ridge regularization.*

16.6 Generalization Error and Early Stopping

We formalize the implicit regularization of GD by analyzing the noisy observation model. Let the training labels be generated by $\mathbf{y}_{\text{train}} = \mathbf{A}\mathbf{x}^* + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

We obtain an estimate $\hat{\mathbf{x}}^{(k)}$ via k iterations of GD on $\mathbf{y}_{\text{train}}$. We wish to analyze the test error (Risk) on a new sample:

$$\mathcal{R}(k) = \frac{1}{n}\mathbb{E}\left\|\mathbf{A}\hat{\mathbf{x}}^{(k)} - \mathbf{y}_{\text{test}}\right\|_2^2 \tag{94}$$

where $\mathbf{y}_{\text{test}} = \mathbf{A}_{\text{test}}\mathbf{x}^* + \boldsymbol{\epsilon}_{test}$ is an independent test label vector with $\mathbf{A}_{\text{test}} \in \mathbb{R}^{n_{\text{test}} \times d}$ and $\boldsymbol{\epsilon}_{test} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_{n_{\text{test}}})$ independent of $\boldsymbol{\epsilon}$. Recall that for the exact OLS estimator $\hat{\mathbf{x}}_{\text{OLS}}$, the risk is strictly $\mathcal{R}(\hat{\mathbf{x}}_{\text{OLS}}) = \sigma^2 + \frac{\sigma^2 d}{n}$. We now track this risk strictly as a function of the iteration count k.

Theorem 16.10 (Noisy Error Dynamics): *Let $\mathbf{A}/\sqrt{n} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ be the thin SVD of the normalized design matrix, implying $\mathbf{H} = \mathbf{V}\boldsymbol{\Sigma}^2\mathbf{V}^\top$. If we project the error into the eigenbasis $\boldsymbol{\alpha}^{(k)} = \mathbf{V}^\top \mathbf{x}^{(k)}$, the error recurrence with noise is:*

$$\boldsymbol{\alpha}^{(k+1)} - \boldsymbol{\alpha}^* = (\mathbf{I} - \eta\boldsymbol{\Sigma}^2)(\boldsymbol{\alpha}^{(k)} - \boldsymbol{\alpha}^*) + \frac{\eta}{\sqrt{n}}\boldsymbol{\Sigma}\tilde{\boldsymbol{\epsilon}} \tag{95}$$

where $\tilde{\boldsymbol{\epsilon}} = \mathbf{U}^\top \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$ is the projected noise vector.

Proof. Substituting the noisy labels into the GD update:

$$\begin{aligned} \mathbf{x}^{(k+1)} &= \mathbf{x}^{(k)} - \eta\left(\mathbf{H}\mathbf{x}^{(k)} - \frac{1}{n}\mathbf{A}^\top \mathbf{y}_{\text{train}}\right) \\ &= \mathbf{x}^{(k)} - \eta\left(\mathbf{H}\mathbf{x}^{(k)} - \frac{1}{n}\mathbf{A}^\top(\mathbf{A}\mathbf{x}^* + \boldsymbol{\epsilon})\right) \\ \mathbf{x}^{(k+1)} - \mathbf{x}^* &= (\mathbf{I} - \eta\mathbf{H})(\mathbf{x}^{(k)} - \mathbf{x}^*) + \frac{\eta}{n}\mathbf{A}^\top\boldsymbol{\epsilon} \end{aligned} \tag{96}$$

Using the thin SVD, the noise term maps to $\frac{\eta}{n}(\sqrt{n}\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^\top)\boldsymbol{\epsilon} = \frac{\eta}{\sqrt{n}}\mathbf{V}\boldsymbol{\Sigma}\tilde{\boldsymbol{\epsilon}}$. Multiplying the entire recurrence by $\mathbf{V}^\top$ to change into the $\boldsymbol{\alpha}$ basis yields:

$$\begin{aligned} \mathbf{V}^\top(\mathbf{x}^{(k+1)} - \mathbf{x}^*) &= \mathbf{V}^\top(\mathbf{I} - \eta\mathbf{V}\boldsymbol{\Sigma}^2\mathbf{V}^\top)(\mathbf{x}^{(k)} - \mathbf{x}^*) + \frac{\eta}{\sqrt{n}}\boldsymbol{\Sigma}\tilde{\boldsymbol{\epsilon}} \\ \boldsymbol{\alpha}^{(k+1)} - \boldsymbol{\alpha}^* &= (\mathbf{I} - \eta\boldsymbol{\Sigma}^2)(\boldsymbol{\alpha}^{(k)} - \boldsymbol{\alpha}^*) + \frac{\eta}{\sqrt{n}}\boldsymbol{\Sigma}\tilde{\boldsymbol{\epsilon}} \end{aligned} \tag{97}$$

Because $\mathbf{U}$ has orthonormal columns ($\mathbf{U}^\top\mathbf{U} = \mathbf{I}_d$), the linear transformation of the isotropic Gaussian noise preserves its distribution: $\tilde{\boldsymbol{\epsilon}} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$. □

Remark 16.11 (The U-Curve and Early Stopping): *The error equation highlights a fundamental bias-variance tradeoff governed by iteration count k:*

- ***Bias (Approximation Error):*** *The term $(\mathbf{I}-\eta\mathbf{\Sigma}^2)^k(\boldsymbol{\alpha}^{(0)}-\boldsymbol{\alpha}^*)$ strictly decays to zero as $k\to\infty$.*
- ***Variance (Noise Accumulation):*** *The term driven by $\tilde{\boldsymbol{\epsilon}}$ is repeatedly accumulated at each iteration. As $k\to\infty$, GD fully memorizes the training noise, and the variance approaches the irreducible limit $\frac{\sigma^2 d}{n}$.*

Consequently, plotting the expected test risk against k produces a characteristic "U-shaped" curve. Terminating the algorithm at an intermediate k^* prior to convergence (Early Stopping) prevents the full accumulation of $\tilde{\boldsymbol{\epsilon}}$, thereby yielding an estimator with lower generalization error than the exact OLS solution.

16.7 Continuous-Time Solution and Expected Risk

We approximate the discrete error dynamics using a continuous model to decouple the effects of initialization bias and stochastic noise.

Lemma 16.12 (Continuous-Time Error Dynamics): *Let $\lambda_i=\sigma_i^2$ be the i-th eigenvalue of $\mathbf{H}$. As the iteration count $k\to t$, the eigen-projected error coordinate $z_i(t)=\alpha_i(t)-\alpha_i^*$ follows the forward-Euler ODE $\dot{z}_i=-\eta\lambda_i z_i+\frac{\eta\sigma_i}{\sqrt{n}}\tilde{\epsilon}_i$. Its solution is,*

$$z_i(t)=\underbrace{e^{-\eta\lambda_i t}z_i(0)}_{\text{transient}}+\underbrace{\frac{1-e^{-\eta\lambda_i t}}{\sqrt{n\lambda_i}}\tilde{\epsilon}_i}_{\text{noise-driven steady state}} \tag{98}$$

Theorem 16.13 (Expected Excess Risk): *Assume the initialization $z_i(0)=\alpha_i^{(0)}-\alpha_i^*$ and the noise $\tilde{\epsilon}_i$ are independent, causing cross-terms to vanish in expectation. The expected excess risk, $\mathcal{R}_{ex}(t)=\sum_{i=1}^d\lambda_i\mathbb{E}[z_i(t)^2]$, decomposes into strictly decreasing bias and strictly increasing variance:*

$$\mathcal{R}_{ex}(t)=\sum_{i=1}^d\underbrace{\lambda_i e^{-2\eta\lambda_i t}(\alpha_i^{(0)}-\alpha_i^*)^2}_{\text{bias}^2}+\sum_{i=1}^d\underbrace{\frac{(1-e^{-\eta\lambda_i t})^2}{n}\sigma^2}_{\text{variance}} \tag{99}$$

Remark 16.14 (Asymptotic Behavior): *As $t\to\infty$, the transient bias decays to 0, and each variance term saturates to σ^2/n. Consequently, $\mathcal{R}_{ex}(\infty)=\frac{\sigma^2 d}{n}$, perfectly recovering the exact OLS risk. The U-shaped trajectory of (99) highlights that terminating the algorithm early inherently regularizes the estimator by preventing full noise accumulation.*

16.8 Equal-Eigenvalue Simplification and Optimal Stopping

To analytically characterize the optimal stopping time t^*, we simplify the landscape curvature.

Theorem 16.15 (Optimal Stopping under Isotropic Curvature): *Assume all eigenvalues are equal ($\lambda_i=\lambda$ for all i) and define the total initialization error as $\mathcal{I}=\left\|\mathbf{x}^{(0)}-\mathbf{x}^*\right\|^2$. The optimal early* *stopping time t^* achieves a minimized excess risk of:*

$$\mathcal{R}_{ex}(t^*) = \frac{\lambda\mathcal{I} \cdot \frac{\sigma^2 d}{n}}{\lambda\mathcal{I} + \frac{\sigma^2 d}{n}} \tag{100}$$

Proof. Under the isotropic assumption $\lambda_i = \lambda$, the excess risk from Theorem 16.13 simplifies to:

$$\mathcal{R}_{\text{ex}}(t) = \lambda e^{-2\eta\lambda t}\mathcal{I} + \frac{\sigma^2 d}{n}(1 - e^{-\eta\lambda t})^2 \tag{101}$$

To find the minimum, we take the derivative with respect to t and set it to zero:

$$\frac{d}{dt}\mathcal{R}_{\text{ex}}(t) = -2\eta\lambda^2 e^{-2\eta\lambda t}\mathcal{I} + 2\eta\lambda\frac{\sigma^2 d}{n}(1 - e^{-\eta\lambda t})e^{-\eta\lambda t} = 0 \tag{102}$$

Dividing out the common strictly positive factor $2\eta\lambda e^{-\eta\lambda t}$ yields:

$$-\lambda e^{-\eta\lambda t}\mathcal{I} + \frac{\sigma^2 d}{n}(1 - e^{-\eta\lambda t}) = 0 \quad \Longrightarrow \quad e^{-\eta\lambda t} = \frac{\frac{\sigma^2 d}{n}}{\lambda\mathcal{I} + \frac{\sigma^2 d}{n}} \tag{103}$$

Substituting this optimal exponential decay factor back into the simplified risk equation directly evaluates to the harmonic mean expression in (100). □

Remark 16.16 (Implicit Regularization Bounds): *The result in (100) takes the form of a harmonic mean: $\frac{ab}{a+b}$. Since $\frac{ab}{a+b} < \min(a, b)$ for all $a, b > 0$, it is strictly guaranteed that $\mathcal{R}_{ex}(t^*)$ is lower than both the starting initialization error ($\lambda\mathcal{I}$) and the converged OLS risk ($\sigma^2 d/n$).*

16.9 Algorithmic Regularization: Early Stopping vs. Ridge Regression

The implicit regularization achieved by terminating Gradient Descent prior to full convergence bears a profound mathematical equivalence to explicit Tikhonov regularization, commonly known as Ridge Regression.

Definition 16.17 (Ridge Estimator): *The Ridge Regression estimator with penalty parameter $\lambda > 0$ is the solution to the heavily penalized empirical risk:*

$$\hat{\mathbf{x}}_\lambda = \arg\min_{\mathbf{x}\in\mathbb{R}^d} \left\{ \frac{1}{2n}\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \frac{\lambda}{2}\|\mathbf{x}\|_2^2 \right\} = (\mathbf{H} + \lambda\mathbf{I})^{-1}\left(\frac{1}{n}\mathbf{A}^\top\mathbf{y}\right) \tag{104}$$

Theorem 16.18 (Spectral Filter Equivalence): *Assuming initialization at the origin ($\mathbf{x}^{(0)} = \mathbf{0}$), Gradient Descent evaluated at iteration k acts as a spectral filter that intimately approximates Ridge Regression with an effective regularization penalty of:*

$$\lambda \approx \frac{1}{\eta k} \tag{105}$$

Proof. Using the exact error operator for GD from Theorem 2 and initializing at $\mathbf{x}^{(0)} = \mathbf{0}$, the k-th iterate $\mathbf{x}^{(k)}$ can be written purely in terms of the optimal unregularized solution $\mathbf{x}^*$:

$$\mathbf{x}^{(k)} = \left(\mathbf{I} - (\mathbf{I} - \eta\mathbf{H})^k\right)\mathbf{x}^* \tag{106}$$

Projecting this onto the eigenbasis of the Hessian $\mathbf{H} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$, the effective scalar multiplier (the "filter factor") applied to the i-th principal component of the solution is:

$$f_{\text{GD}}^{(k)}(\lambda_i) = 1 - (1 - \eta\lambda_i)^k \tag{107}$$

Conversely, for Ridge Regression, we rewrite the estimator as $\hat{\mathbf{x}}_\lambda = (\mathbf{H} + \lambda\mathbf{I})^{-1}\mathbf{H}\mathbf{x}^*$. In the same eigenbasis, its filter factor is:

$$f_{\text{Ridge}}^{\lambda}(\lambda_i) = \frac{\lambda_i}{\lambda_i + \lambda} \tag{108}$$

To link these, we use the continuous-time approximation for GD (assuming small step size η and large k). We apply the exponential limit $(1 - \eta\lambda_i)^k \approx e^{-\eta k \lambda_i}$. Equating the two continuous-time filter factors yields:

$$1 - e^{-\eta k \lambda_i} \approx \frac{\lambda_i}{\lambda_i + \lambda} \tag{109}$$

By taking the first-order Taylor expansion of the exponential term for small eigenvalues (where regularization exerts the most critical influence, meaning $\lambda_i \to 0$), we obtain $1 - e^{-\eta k \lambda_i} \approx \eta k \lambda_i$. Equating this local approximation to the Ridge filter yields:

$$\eta k \lambda_i \approx \frac{\lambda_i}{\lambda_i + \lambda} \implies \eta k(\lambda_i + \lambda) \approx 1 \tag{110}$$

For heavily penalized small eigenvalues where $\lambda_i \ll \lambda$, this simplifies strictly to $\eta k \lambda \approx 1$, producing the inverse relationship $\lambda \approx \frac{1}{\eta k}$. □

16.10 Introduction to Line Search

So far we have used a *fixed* step size η for all iterations. A natural improvement is to allow η_k to change adaptively at each step.

Exact Line Search

At iteration k, given the current point $\mathbf{x}^{(k)}$ and descent direction $\mathbf{p}^{(k)} = -\nabla f(\mathbf{x}^{(k)})$, choose:

$$\eta_k = \arg\min_{s \geq 0} f(\mathbf{x}^{(k)} + s\mathbf{p}^{(k)}) \tag{111}$$

This minimizes the objective along the ray from $\mathbf{x}^{(k)}$ in direction $\mathbf{p}^{(k)}$—hence the name *line search.*

Remark 16.19 (Solving the sub-problem): *The exact line search equation is a one-dimensional optimization in s. For least squares, the objective along the ray is quadratic in s, so the minimizer is found in closed form by setting the derivative to zero. For general smooth functions, a scalar root-finding method may be needed.*

Remark 16.20 (In practice): *For least squares, exact line search does not dramatically improve the convergence rate over a well-chosen fixed step size. In practice, however, line search often helps, especially when the landscape is poorly approximated by a quadratic.*

17 Stochastic Gradient Descent (SGD)

TODO: second half of lecture 10.2 and lecture 10.3 d

18 Gradient Descent For Beyond Least Squares

Maps to lecture 11, 11.1, 11.2, and 11.3.

19 Subgradient Descent

20 Alternating Projections

21 Proximal Methods